Math 231 – Linear Algebra: Quiz, Sections 4.1–4.4

Detailed marking rubric and answer keys, versions A–J

Course: Math 231 – Linear Algebra, Qatar University, Department of Mathematics, Statistics and Physics. **Instructor:** Dr. Yousef Dabboorasad. **Term:** Fall 2026.

Scope: Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence). **Maximum marks:** 10 per version. **Authorized source:** the Math 231 Overleaf project notes, file `math231_lecture_notes_v02_2026_fully_solved.tex`, lines 10672–13885 (Chapter 4, Sections 4.1–4.4), and the Chapter 4 Beamer deck.

1. Structure and marking conventions

- **Problem 1 (True/False, 4 statements × 0.5 = 2 marks).** One statement from each of Sections 4.1, 4.2, 4.3, 4.4. Full credit for a single clearly ticked correct box; zero for a blank, for two ticks, or for the wrong box. No justification is required and none is marked.
- **Problem 2 (Multiple choice, 4 questions × 1 = 4 marks).** (a) Section 4.1: identify the single failing vector-space axiom; (b) Section 4.2: Subspace Test; (c) Section 4.3: membership in a span; (d) Section 4.4: Wronskian computation and interpretation. Exactly one option is correct; a question with more than one option ticked receives zero (course policy, as on the midterm).
- **Problem 3 (written, 4 marks).** Part (a) linear independence/dependence of three vectors (polynomials) [2.5]; part (b) membership of a given vector in $\operatorname{span}(S)$ [1.5]. Marked with the eight evidence checkpoints a1–a5, b1–b3 below (same structure and maxima in every version). Any mathematically valid method within Sections 1.1–2.3 and 4.1–4.4 (row reduction, determinant criterion, Cramer's rule, an explicit verified dependence relation) earns the corresponding checkpoints.
- **Follow-through.** An arithmetic slip is penalized once, at the checkpoint where it occurs; later checkpoints are awarded if the subsequent logic is valid for the (incorrect) numbers obtained. A bare final classification with no supporting work earns a5 (or b3) only when the stated reason is itself valid (e.g. an explicit nontrivial relation).
- **Equivalent forms.** Accept vectors written as rows or columns, any order of elementary row operations, REF or RREF, and the polynomial/coefficient-vector identification $a + bx + cx^2 \leftrightarrow (a, b, c)$.
- **Design note.** In every version the written problem has the same two-part structure; versions A, D, E, H, I have a linearly dependent S with $\mathbf{u} \notin \operatorname{span}(S)$, and versions B, C, F, G, J have a linearly independent S with $\mathbf{u} \in \operatorname{span}(S)$ (unique integer coefficients). Workload is comparable: a 3×3 elimination in both parts, with back-substitution (independent versions) or an inconsistency check (dependent versions).

2. Master answer grid

Ver.	1(a)	1(b)	1(c)	1(d)	2(a)	2(b)	2(c)	2(d)	3(a)	3(b)
A	T	F	T	F	B	C	B	B	dependent	no: $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent row)
B	F	T	F	T	C	B	C	D	independent	yes: $(k_1, k_2, k_3) = (-1, 2, 1)$
C	T	F	F	T	A	D	A	A	independent	yes: $(k_1, k_2, k_3) = (2, -1, 1)$
D	F	T	T	F	D	A	D	C	dependent	no: $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent row)
E	T	T	F	F	B	C	B	A	dependent	no: $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent row)
F	F	F	T	T	D	B	C	D	independent	yes: $(k_1, k_2, k_3) = (1, 2, -2)$
G	T	F	T	F	C	D	D	B	independent	yes: $(k_1, k_2, k_3) = (1, 2, -1)$
H	F	T	F	T	A	A	A	C	dependent	no: $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent row)
I	F	T	T	F	C	D	C	B	dependent	no: $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent row)
J	T	F	F	T	B	B	B	A	independent	yes: $(k_1, k_2, k_3) = (-1, 1, 2)$

3. Checkpoints for Problem 3 (identical in all versions)

Check	Marks	Required evidence	Accepted alternatives / notes
a1	0.5	Sets up the vector equation $k_1\mathbf{v}_1+k_2\mathbf{v}_2+k_3\mathbf{v}_3=\mathbf{0}$ (for polynomials: equates coefficients of $1, x, x^2$) and states that the question is whether only the trivial solution exists.	Writing the homogeneous augmented matrix directly, or stating the determinant criterion for three vectors in a 3-dimensional space, is accepted.
a2	0.5	Writes the correct homogeneous linear system (three equations) or the correct coefficient/augmented matrix $[A \mid \mathbf{0}]$.	Transposed set-up (vectors as rows) is accepted provided the subsequent reasoning is consistent.
a3	0.5	First elimination stage correct: zeros below the first pivot with correct row operations.	Any valid sequence of elementary row operations; a determinant expansion with correct cofactors earns a3+a4 together.
a4	0.5	Completes the reduction to a row echelon form correctly (identifies the zero row, or the third pivot) or computes $\det(A)$ correctly.	One arithmetic slip that does not change the structural outcome: deduct here only, follow through in a5.
a5	0.5	Correct conclusion with a valid reason: a free variable / nontrivial solutions $\Rightarrow$ linearly dependent, or only the trivial solution $\Rightarrow$ linearly independent (or $\det = 0$ / $\det \neq 0$ respectively).	Exhibiting an explicit nontrivial relation such as $\mathbf{v}_3 = \mathbf{v}_1 + 2\mathbf{v}_2$ proves dependence and earns a5 (and a1–a4 if the relation is verified by computation).
b1	0.5	Sets up $k_1\mathbf{v}_1+k_2\mathbf{v}_2+k_3\mathbf{v}_3=\mathbf{u}$ as a linear system or augmented matrix $[A \mid \mathbf{u}]$.	Re-using the matrix from part (a) with the new right-hand side is accepted and encouraged.
b2	0.5	Row reduces $[A \mid \mathbf{u}]$ correctly (or applies the same operations as in (a) to the new column) to a row echelon form.	If (a) was reduced incorrectly, award b2 for consistent application of the (wrong) operations only when the resulting logic is sound; otherwise follow through once.
b3	0.5	Correct final answer: inconsistent system $\Rightarrow$ $\mathbf{u} \notin \operatorname{span}(S)$; or consistent $\Rightarrow$ $\mathbf{u} \in \operatorname{span}(S)$ with correct k_1, k_2, k_3 and the combination written out.	For the independent versions, k_1, k_2, k_3 may also be found by Cramer's rule or the inverse matrix (Chapter 2); the written combination $\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3$ must be stated.
Total	4		

4. Notes on the source (instructor information)

- The A4 notes' solution to **Example 4.4.3** (line 13467) is a copy of the Example 4.4.4 solution: it speaks of "$r = 5$ polynomials" and "Fact 4.4.1(2)" although the example has four vectors in $\mathbb{R}^3$; the correct argument is $r = 4 > n = 3$ by Fact 4.4.1(1), as printed in the Beamer deck. No quiz item depends on the erroneous text.
- In **Extra Problems 4.2.7** (line 12202) the sentence "The set is linearly dependent (4 equations, 3 unknowns, consistent with a zero row)" is not justified by the computation: the coefficient matrix has three pivots, so $\{M_1, M_2, M_3\}$ is linearly independent; the zero row only reflects that four equations were used for three unknowns. The quiz does not use this remark.
- **Extra Problems 4.3.2 and 4.3.4** (lines 12992, 13115) are the same problem with the same set S. Both were treated as one source item.
- The notes write P_n in Sections 4.2–4.3 and $\mathbb{P}_n$ in Section 4.4; the quiz uses P_n throughout, as in the definition box (line 11369).

Version A

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	In any vector space V, $0\mathbf{u} = \mathbf{0}$ for every vector $\mathbf{u}$ in V.	**TRUE**	This is Theorem 4.1.1(a). Source: Theorem 4.1.1, line 10790
(b)	The set of all invertible 2×2 matrices is a subspace of M_{22}.	**FALSE**	The zero matrix is not invertible, so condition (a) of the Subspace Test fails (Example 4.2.4). Source: Example 4.2.4 (Invertible Matrices Do NOT Form a Subspace), line 11293; Theorem 4.2.1 (The Subspace Test), line 11139
(c)	If $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_r\}$ is a set of vectors in a vector space V, then $\text{span}(S)$ is a subspace of V.	**TRUE**	Theorem 4.3.1. Source: Theorem 4.3.1 (Span Is a Subspace), line 12627
(d)	If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, then $\mathbf{v}_3$ must be a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.	**FALSE**	The Remark after Theorem 4.4.1 says *at least one* vector is a linear combination of the others, not necessarily $\mathbf{v}_3$: for $\mathbf{v}_1 = (1,0,0)$, $\mathbf{v}_2 = (2,0,0)$, $\mathbf{v}_3 = (0,1,0)$ the set is dependent but $\mathbf{v}_3 \neq k_1\mathbf{v}_1 + k_2\mathbf{v}_2$. Source: Remark after Theorem 4.4.1, line 13267; Theorem 4.4.1, line 13253

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (kx_1,\ 0)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

B* Axiom 10: $1\mathbf{u} = \mathbf{u}$

C Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

D Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

Key: B (Axiom 10). Take $\mathbf{u} = (7, 8)$: $1\mathbf{u} = (1 \cdot 7,\ 0) = (7, 0) \neq (7, 8) = \mathbf{u}$, so Axiom 10 fails. The usual addition makes Axioms 7, 8, 9 hold: e.g. $(k + m)\mathbf{u} = ((k + m)u_1, 0) = (ku_1, 0) + (mu_1, 0) = k\mathbf{u} + m\mathbf{u}$.

Source: Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1, x_2) = (kx_1, x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Identical operations to Example 4.1.3 (the notes' own example); re-set as a four-option item.

(b) Which of the following subsets W is a **subspace** of $V = \mathbb{R}^2$ (with the usual operations)?

A $W = \{(x, y) : x + y = 1\}$ — **not a subspace.** $(0, 0) \notin W$ since $0 + 0 \neq 1$ (condition (a) fails).

B $W = \{(x, y) : y = x^2\}$ — **not a subspace.** $(1, 1) \in W$ but $2(1, 1) = (2, 2) \notin W$ since $2 \neq 2^2$ (condition (c) fails).

C* $W = \{(x, y) : y = 3x\}$ — **subspace.** $(0, 0) \in W$; if $y_1 = 3x_1$, $y_2 = 3x_2$ then $y_1 + y_2 = 3(x_1 + x_2)$ and $ky_1 = 3(kx_1)$, so W is closed under both operations.

D $W = \{(x, y) : x \geq 0,\ y \geq 0\}$ — **not a subspace.** $(5, 7) \in W$ but $(-1)(5, 7) = (-5, -7) \notin W$ (Example 4.2.2).

Key: C. Source: Theorem 4.2.1 (The Subspace Test), line 11139; Example 4.2.2 (First Quadrant Is NOT a Subspace), line 11200; Remark 4.2.1 (Showing a Set Is NOT a Subspace), line 11154. Adaptation: Line through the origin and three non-subspaces modelled on Example 4.2.2 / Remark 4.2.1.

(c) Let $\mathbf{v}_1 = (1, 0, 2)$ and $\mathbf{v}_2 = (0, 1, -1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

A $(3, 1, 7)$

B* $(3, 1, 5)$

C $(1, 1, 2)$

D $(2, 2, 0)$

Key: B. Write $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ and match the candidate: two of the three components determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = 3$, $k_2 = 1$ would force the third component to be $5 \neq 7$. **B**: $k_1 = 3$, $k_2 = 1$ give third component $= 5$ ✓, so $(3, 1, 5) = 3\mathbf{v}_1 + \mathbf{v}_2$. C: $k_1 = 1$, $k_2 = 1$ would force the third component to be $1 \neq 2$. D: $k_1 = 2$, $k_2 = 2$ would force the third component to be $2 \neq 0$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = x$ and $f_2 = e^x$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A $W(x) = (x - 1)e^x$; the set is linearly dependent because $W(1) = 0$

B* $W(x) = (x - 1)e^x$; the set is linearly independent

C $W(x) = (x + 1)e^x$; the set is linearly independent

D $W(x) = 0$ for all x; the Wronskian test is inconclusive

Key: B. $W(x) = \begin{vmatrix} x & e^x \\ 1 & e^x \end{vmatrix} = xe^x - e^x = (x - 1)e^x$. This is not identically zero (e.g. $W(0) = -1$), so by Fact 4.4.2 the set is linearly independent. $W(1) = 0$ at a single point does not matter (Remark, item 2); $(x + 1)e^x$ is a sign error; W is not the zero function.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 2, -1), \qquad \mathbf{v}_2 = (2, 1, 1), \qquad \mathbf{v}_3 = (5, 4, 1),$$

and let $\mathbf{u} = (1, 1, 3)$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1, 2, -1) + k_2(2, 1, 1) + k_3(5, 4, 1) = (0, 0, 0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + 2k_2 + 5k_3 = 0 \\ 2k_1 + k_2 + 4k_3 = 0 \\ -k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 2 & 1 & 4 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow[R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 3 & 6 & 0 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = -2$ and $k_1 = -1$, so $-\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, i.e. $\mathbf{v}_3 = \mathbf{v}_1 + 2\mathbf{v}_2$.

S is linearly dependent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (1, 1, 3)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 2 & 5 & 1 \\ 2 & 1 & 4 & 1 \\ -1 & 1 & 1 & 3 \end{array}\right] \xrightarrow[R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 1 \\ 0 & -3 & -6 & -1 \\ 0 & 3 & 6 & 4 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 5 & 1 \\ 0 & -3 & -6 & -1 \\ 0 & 0 & 0 & 3 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 3$, i.e. $0 = 3$, which is impossible: the system is **inconsistent**. Hence $\mathbf{u}$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

$\boxed{\mathbf{u} \notin \operatorname{span}(S).}$

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side $\mathbf{0}$.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 2 & 5 & 0 \\ 0 & -3 & -6 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) = 0$.
a5	0.5	Conclusion **linearly dependent** with reason (zero row / free variable / nontrivial solution / det = 0). An explicit verified relation such as $\mathbf{v}_3 = \mathbf{v}_1 + 2\mathbf{v}_2$ is a complete justification.
b1	0.5	System or augmented matrix with right-hand side $(1, 1, 3)$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 2 & 5 & 1 \\ 0 & -3 & -6 & -1 \\ 0 & 0 & 0 & 3 \end{array}\right]$.
b3	0.5	States that the last row is impossible ($0 = 3$), hence $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent: $0 = 3$). A correct argument via the dependence relation (e.g. that $\operatorname{span}(S) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathbf{u}$ is not a combination of two vectors) is also accepted.
Total	4	

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version B

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	A set with two operations that satisfies nine of the ten vector space axioms is still a vector space.	**FALSE**	All ten axioms must hold (Definition 4.1.1); if even one axiom fails the set is not a vector space (Examples 4.1.3 and 4.1.4 fail a single axiom). Source: Definition 4.1.1 (Vector Space Axioms), line 10737; Example 4.1.3 (NOT a vector space: $k(x_1,x_2)=(kx_1,0)$), line 10977
(b)	If A is an $m\times n$ matrix, then the solution set of the homogeneous system $A\mathbf{x}=\mathbf{0}$ is a subspace of $\mathbb{R}^n$.	**TRUE**	Example 4.2.7: $A\mathbf{0}=\mathbf{0}$, $A(\mathbf{u}+\mathbf{v})=\mathbf{0}$ and $A(k\mathbf{u})=\mathbf{0}$. Source: Example 4.2.7 (Solution Spaces of Homogeneous Systems), line 11477
(c)	$\operatorname{span}\{1,\ x,\ x^2\}=P_3$.	**FALSE**	Every linear combination $k_1+k_2x+k_3x^2$ has no x^3 term, so $x^3\in P_3$ is not in the span; the span is P_2 (Remark 4.3.2). Source: Remark 4.3.2 (Important Spanning Sets), line 12664; Definition 4.3.1 (Span), line 12614
(d)	Every set of four vectors in $\mathbb{R}^3$ is linearly dependent.	**TRUE**	Fact 4.4.1(1) with $r=4>n=3$ (Example 4.4.3). Source: Fact 4.4.1 (Dimension Argument for Linear Dependence), line 13452; Example 4.4.3 (Dimension Argument in $\mathbb{R}^3$), line 13467

Problem 2 – Multiple choice (1 mark each)

(a) Let $V=\mathbb{R}^2$ with the usual addition $(x_1,x_2)+(y_1,y_2)=(x_1+y_1,\ x_2+y_2)$ and the scalar multiplication $k(x_1,x_2)=(kx_1,\ x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 10: $1\mathbf{u}=\mathbf{u}$

B Axiom 7: $k(\mathbf{u}+\mathbf{v})=k\mathbf{u}+k\mathbf{v}$

C* Axiom 8: $(k+m)\mathbf{u}=k\mathbf{u}+m\mathbf{u}$

D Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

Key: C (Axiom 8). Take $\mathbf{u}=(2,5)$, $k=2$, $m=3$: $(k+m)\mathbf{u}=5(2,5)=(10,5)$ but $k\mathbf{u}+m\mathbf{u}=(4,5)+(6,5)=(10,10)$, so Axiom 8 fails. Axiom 10: $1\mathbf{u}=(u_1,u_2)=\mathbf{u}$ holds; Axiom 7: $k(\mathbf{u}+\mathbf{v})=(k(u_1+v_1),u_2+v_2)=k\mathbf{u}+k\mathbf{v}$ holds; Axiom 6 holds since $(ku_1,u_2)\in\mathbb{R}^2$.

Source: Example 4.1.3 (NOT a vector space: $k(x_1,x_2)=(kx_1,0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1,x_2)=(kx_1,x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Identical operations to Example 4.1.4 (the notes' own example); re-set as a four-option item.

(b) Which of the following subsets of $V=M_{22}$ (with the usual operations) is **not** a subspace of V?

A all symmetric matrices ($A^T=A$) — **subspace.** Example 4.2.3: $O^T=O$, $(A+B)^T=A+B$, $(kA)^T=kA$.

B* all invertible matrices — **not a subspace.** The zero matrix is not invertible, so $O\notin W$ (Example 4.2.4).

C all matrices with $\operatorname{tr}(A)=0$ — **subspace.** Extra Problems 4.2.6: $\operatorname{tr}(O)=0$, $\operatorname{tr}(A+B)=\operatorname{tr}A+\operatorname{tr}B=0$, $\operatorname{tr}(kA)=k\operatorname{tr}A=0$.

D all matrices $\begin{bmatrix}a & b\\ c & d\end{bmatrix}$ with $a=0$ — **subspace.** O has $a=0$; the $(1,1)$ entry of $A+B$ is $0+0=0$ and of kA is $k\cdot 0=0$.

Key: B. Source: Example 4.2.3 (Symmetric Matrices Form a Subspace), line 11245; Example 4.2.4 (Invertible Matrices Do NOT Form a Subspace), line 11293; Extra Problems 4.2.6 (trace zero), line 12157; Theorem 4.2.1 (The Subspace Test), line 11139. Adaptation: Three subspaces from the notes' examples plus the non-subspace of Example 4.2.4.

(c) Let $p_1 = 1 + x^2$ and $p_2 = x$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

A $3 + 2x - 3x^2$

B $2 + 3x + 3x^2$

C* $3 + 2x + 3x^2$

D $3 + 3x + 2x^2$

Key: C. Write $k_1 p_1 + k_2 p_2$ and match the candidate: two of the three coefficients determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = 3$, $k_2 = 2$ would force the x^2 coefficient to be $3 \neq -3$. B: $k_1 = 2$, $k_2 = 3$ would force the x^2 coefficient to be $2 \neq 3$. **C**: $k_1 = 3$, $k_2 = 2$ give x^2 coefficient $= 3$ ✓, so $3 + 2x + 3x^2 = 3p_1 + 2p_2$. D: $k_1 = 3$, $k_2 = 3$ would force the x^2 coefficient to be $3 \neq 2$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = e^x$ and $f_2 = e^{2x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A $W(x) = 3e^{3x}$; the set is linearly independent

B $W(x) = 0$ for all x; the Wronskian test is inconclusive

C $W(x) = e^{3x}$; the set is linearly dependent because e^{2x} is a multiple of e^x

D* $W(x) = e^{3x}$; the set is linearly independent

Key: D. $W(x) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{3x} - e^{3x} = e^{3x} \neq 0$ for every x, so the set is linearly independent (Fact 4.4.2; compare Example 4.4.5). $3e^{3x}$ is an arithmetic error; $e^{2x} = e^x \cdot e^x$ is not a *scalar* multiple of e^x; W is never zero.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x + x^2, \qquad p_2 = 1 - x + 2x^2, \qquad p_3 = 2 + x - x^2,$$

and let $u = 3 - 2x + 2x^2$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether u belongs to $\text{span}(S)$, that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1 p_1 + k_2 p_2 + k_3 p_3 = 0$, i.e. $k_1(1 + x + x^2) + k_2(1 - x + 2x^2) + k_3(2 + x - x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + 2k_3 = 0 \\ k_1 - k_2 + k_3 = 0 \\ k_1 + 2k_2 - k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & -1 & 1 & 0 \\ 1 & 2 & -1 & 0 \end{array}\right] \xrightarrow[-R_1+R_3 \to R_3]{-R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 1 & -3 & 0 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & -2 & -1 & 0 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -7 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 3 - 2x + 2x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 1 & -1 & 1 & -2 \\ 1 & 2 & -1 & 2 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & -2 & -1 & -5 \\ 0 & 1 & -3 & -1 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & -2 & -1 & -5 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & -7 & -7 \end{array}\right]$$

Back-substitution: $-7k_3 = -7 \Rightarrow k_3 = 1$; $k_2 - 3k_3 = -1 \Rightarrow k_2 = 2$; $k_1 + k_2 + 2k_3 = 3 \Rightarrow k_1 = -1$. Hence $u = -p_1 + 2p_2 + p_3$.

$u \in \mathrm{span}(S)$: $u = -p_1 + 2p_2 + p_3$.

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side 0.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 2 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -7 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) \neq 0$ computed correctly.
a5	0.5	Conclusion **linearly independent** with reason (three pivots / only the trivial solution / $\det \neq 0$).
b1	0.5	System or augmented matrix with right-hand side $3 - 2x + 2x^2$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 2 & 3 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & -7 & -7 \end{array}\right]$.
b3	0.5	Correct coefficients and the combination written out: $u = -p_1 + 2p_2 + p_3$. Cramer's rule / inverse matrix accepted.
Total	4	

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version C

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	If $\mathbf{u}$ is a vector in a vector space and k is a scalar with $k\mathbf{u} = \mathbf{0}$, then $k = 0$ or $\mathbf{u} = \mathbf{0}$.	**TRUE**	This is Theorem 4.1.1(d). Source: Theorem 4.1.1, line 10790
(b)	If W_1 and W_2 are subspaces of a vector space V, then $W_1 \cup W_2$ is always a subspace of V.	**FALSE**	Remark 4.2.2 / Example 4.2.9: the union of the x-axis and the y-axis is not closed under addition, $(0,7) + (2,0) = (2,7)$. Source: Remark 4.2.2 (Union of Subspaces), line 11677; Example 4.2.9 (Union of Subspaces Is NOT a Subspace), line 11687
(c)	The vectors $(1,0,0)$ and $(0,1,0)$ span $\mathbb{R}^3$.	**FALSE**	$k_1(1,0,0) + k_2(0,1,0) = (k_1, k_2, 0)$ always has third component 0, so e.g. $(0,0,1)$ is not a linear combination (Definition 4.3.1). Source: Definition 4.3.1 (Span), line 12614; Example 4.3.1 (Testing Whether Vectors Span $\mathbb{R}^3$), line 12721
(d)	A set with exactly two vectors is linearly independent if and only if neither vector is a scalar multiple of the other.	**TRUE**	Theorem 4.4.1(c). Source: Theorem 4.4.1, line 13253

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (k^2 x_1,\ k^2 x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A* Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

B Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

C Axiom 10: $1\mathbf{u} = \mathbf{u}$

D Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

Key: A (Axiom 8). Take $\mathbf{u} = (1,1)$, $k = m = 1$: $(k + m)\mathbf{u} = 2(1,1) = (4,4)$ but $k\mathbf{u} + m\mathbf{u} = (1,1) + (1,1) = (2,2)$, so Axiom 8 fails. Axiom 9: $k(m\mathbf{u}) = k^2 m^2 \mathbf{u} = (km)^2\mathbf{u} = (km)\mathbf{u}$ holds; Axiom 10: $1^2\mathbf{u} = \mathbf{u}$ holds; Axiom 7: $k^2(\mathbf{u} + \mathbf{v}) = k^2\mathbf{u} + k^2\mathbf{v}$ holds.

Source: Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1, x_2) = (kx_1, x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: New scalar multiplication in the style of Examples 4.1.3-4.1.4; only Axiom 8 fails.

(b) Which of the following subsets W is a **subspace** of $V = P_2$ (with the usual operations)?

A $W = \{p \in P_2 : \deg(p) = 2\}$ — **not a subspace.** The zero polynomial does not have degree 2, so $\mathbf{0} \notin W$ (Example 4.2.6).

B $W = \{a + bx + cx^2 : a, b, c \text{ are integers}\}$ — **not a subspace.** $1 + x + x^2 \in W$ but $\frac{1}{2}(1 + x + x^2) \notin W$ (Extra Problems 4.2.4).

C $W = \{p \in P_2 : p(0) = 1\}$ — **not a subspace.** The zero polynomial has $p(0) = 0 \neq 1$, so $\mathbf{0} \notin W$.

D* $W = \{a + bx + cx^2 : a + b + c = 0\}$ — **subspace.** Extra Problems 4.2.2: $0 + 0 + 0 = 0$, and the coefficient sum of $p_1 + p_2$ and of kp is 0.

Key: D. Source: Extra Problems 4.2.2 ($a + b + c = 0$ in P_2), line 12017; Example 4.2.6 (Degree Exactly 2 Is NOT a Subspace), line 11424; Extra Problems 4.2.4 (integer coefficients), line 12098; Theorem 4.2.1 (The Subspace Test), line 11139. Adaptation: Direct use of the notes' P_2 examples; $p(0) = 1$ added as a non-subspace.

(c) Let $\mathbf{v}_1 = (1, 0, -1)$ and $\mathbf{v}_2 = (0, 1, 3)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

A* $(2, 1, 1)$

B $(2, 1, 5)$

C $(1, 2, -5)$

D $(2, 1, -1)$

Key: A. Write $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ and match the candidate: two of the three components determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). **A**: $k_1 = 2$, $k_2 = 1$ give third component = 1 ✓, so $(2, 1, 1) = 2\mathbf{v}_1 + \mathbf{v}_2$. B: $k_1 = 2$, $k_2 = 1$ would force the third component to be $1 \neq 5$. C: $k_1 = 1$, $k_2 = 2$ would force the third component to be $5 \neq -5$. D: $k_1 = 2$, $k_2 = 1$ would force the third component to be $1 \neq -1$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = 1$ and $f_2 = e^{-x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A* $W(x) = -e^{-x}$; the set is linearly independent

B $W(x) = e^{-x}$; the set is linearly independent

C $W(x) = -e^{-x}$; the set is linearly dependent because $W(x) < 0$

D $W(x) = 0$ for all x; the Wronskian test is inconclusive

Key: A. $W(x) = \begin{vmatrix} 1 & e^{-x} \\ 0 & -e^{-x} \end{vmatrix} = -e^{-x} \neq 0$ for every x, hence linearly independent (Fact 4.4.2). The sign of W is irrelevant; only $W \equiv 0$ would be inconclusive; e^{-x} is a sign error in the derivative.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 1, 2), \qquad \mathbf{v}_2 = (1, 0, 1), \qquad \mathbf{v}_3 = (2, 1, 1),$$

and let $\mathbf{u} = (3, 3, 4)$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1, 1, 2) + k_2(1, 0, 1) + k_3(2, 1, 1) = (0, 0, 0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + 2k_3 = 0 \\ k_1 + k_3 = 0 \\ 2k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow[-2R_1+R_3 \to R_3]{-R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -3 & 0 \end{array}\right] \xrightarrow{-R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (3, 3, 4)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 3 \\ 2 & 1 & 1 & 4 \end{array}\right] \xrightarrow[-2R_1+R_3\to R_3]{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -3 & -2 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -2 & -2 \end{array}\right]$$

Back-substitution: $-2k_3 = -2 \Rightarrow k_3 = 1$; $-k_2 - k_3 = 0 \Rightarrow k_2 = -1$; $k_1 + k_2 + 2k_3 = 3 \Rightarrow k_1 = 2$. Hence $\mathbf{u} = 2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3$.

$\mathbf{u} \in \text{span}(S)$: $\mathbf{u} = 2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3$.

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side $\mathbf{0}$.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) \neq 0$ computed correctly.
a5	0.5	Conclusion **linearly independent** with reason (three pivots / only the trivial solution / $\det \neq 0$).
b1	0.5	System or augmented matrix with right-hand side $(3, 3, 4)$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 2 & 3 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & -2 & -2 \end{array}\right]$.
b3	0.5	Correct coefficients and the combination written out: $\mathbf{u} = 2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3$. Cramer's rule / inverse matrix accepted.
Total	4	

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version D

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	In a vector space, if $k\mathbf{u} = \mathbf{0}$ for a scalar k and a vector $\mathbf{u}$, then both $k = 0$ and $\mathbf{u} = \mathbf{0}$.	**FALSE**	Theorem 4.1.1(d) gives $k = 0$ *or* $\mathbf{u} = \mathbf{0}$, not both: e.g. $0\mathbf{u} = \mathbf{0}$ for $\mathbf{u} = (1, 1) \neq \mathbf{0}$. Source: Theorem 4.1.1, line 10790
(b)	If W_1 and W_2 are subspaces of a vector space V, then $W_1 \cap W_2$ is a subspace of V.	**TRUE**	Theorem 4.2.2 (intersection of subspaces). Source: Theorem 4.2.2 (Intersection of Subspaces), line 11551
(c)	Every vector in a set S belongs to span(S).	**TRUE**	Remark 4.3.1(1): $\mathbf{w}_i = 0\mathbf{w}_1 + \cdots + 1\mathbf{w}_i + \cdots + 0\mathbf{w}_r$. Source: Remark 4.3.1 (Properties of Span), line 12637
(d)	The functions e^x and $6e^x$ form a linearly independent set in $C(-\infty, \infty)$.	**FALSE**	$6e^x$ is a scalar multiple of e^x, so the set is linearly dependent (Example 4.4.7, Theorem 4.4.1(c)). Source: Example 4.4.7 (Wronskian with Scalar Multiples), line 13707; Theorem 4.4.1, line 13253

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (0,\ kx_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

B Axiom 4: there is a zero vector $\mathbf{0}$ with $\mathbf{0} + \mathbf{u} = \mathbf{u}$

C Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

D* Axiom 10: $1\mathbf{u} = \mathbf{u}$

Key: D (Axiom 10). Take $\mathbf{u} = (3, 4)$: $1\mathbf{u} = (0, 4) \neq (3, 4) = \mathbf{u}$, so Axiom 10 fails. Axiom 6 holds since $(0, ku_2) \in \mathbb{R}^2$; Axiom 4 holds with $\mathbf{0} = (0, 0)$ (usual addition); Axiom 7: $k(\mathbf{u} + \mathbf{v}) = (0, k(u_2 + v_2)) = (0, ku_2) + (0, kv_2)$ holds.

Source: Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1, x_2) = (kx_1, x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Mirror image of Example 4.1.3 (the zero is placed in the first component).

(b) Which of the following subsets of $V = \mathbb{R}^3$ (with the usual operations) is **not** a subspace of V?

A* $W = \{(x, y, z) : x + y + z = 1\}$ — **not a subspace.** $(0, 0, 0) \notin W$ since $0 \neq 1$ (condition (a) fails; compare Example 4.2.8).

B $W = \{(x, y, z) : x + y + z = 0\}$ — **subspace.** Solution space of the homogeneous equation $x + y + z = 0$ (Example 4.2.7).

C $W = \{(x, y, z) : x = y = z\}$ — **subspace.** Solution space of the homogeneous system $x - y = 0,\ y - z = 0$ (Example 4.2.7).

D $W = \{(x, y, z) : x = 2y \text{ and } z = 0\}$ — **subspace.** Solution space of the homogeneous system $x - 2y = 0,\ z = 0$ (Example 4.2.7).

Key: A. Source: Example 4.2.7 (Solution Spaces of Homogeneous Systems), line 11477; Example 4.2.8 (Solution Set of a Non-Homogeneous System), line 11525; Theorem 4.2.1 (The Subspace Test), line 11139. Adaptation: Homogeneous versus non-homogeneous solution sets (Examples 4.2.7-4.2.8) written as explicit subsets of $\mathbb{R}^3$.

(c) Let $p_1 = 1 + x$ and $p_2 = x^2$ in P_2. Which of the following polynomials belongs to span$\{p_1, p_2\}$?

A $2 - 2x - x^2$

B $2 + x - x^2$

C $1 + 2x + 2x^2$

D* $2 + 2x - x^2$

Key: D. Write $k_1p_1 + k_2p_2$ and match the candidate: two of the three coefficients determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = 2$, $k_2 = -1$ would force the x coefficient to be $2 \neq -2$. B: $k_1 = 2$, $k_2 = -1$ would force the x coefficient to be $2 \neq 1$. C: $k_1 = 1$, $k_2 = 2$ would force the x coefficient to be $1 \neq 2$. **D**: $k_1 = 2$, $k_2 = -1$ give x coefficient $= 2$ ✓, so $2 + 2x - x^2 = 2p_1 - p_2$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = x$ and $f_2 = x^2$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A $W(x) = x^2$; the set is linearly dependent because $W(0) = 0$

B $W(x) = x^2$; the Wronskian test is inconclusive because $W(0) = 0$

C* $W(x) = x^2$; the set is linearly independent

D $W(x) = 3x^2$; the set is linearly independent

Key: C. $W(x) = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x^2 - x^2 = x^2$. Since $W(1) = 1 \neq 0$, W is not identically zero and the set is linearly independent (Fact 4.4.2). $W(0) = 0$ at one point neither proves dependence nor makes the test inconclusive (Remark, item 2); $3x^2$ is an arithmetic error.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x, \qquad p_2 = 1 - x + x^2, \qquad p_3 = 1 + 3x - x^2,$$

and let $u = 1 + 2x + x^2$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(1+x) + k_2(1-x+x^2) + k_3(1+3x-x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + k_3 = 0 \\ k_1 - k_2 + 3k_3 = 0 \\ k_2 - k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & -1 & 3 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \xrightarrow{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -2 & 2 & 0 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = 1$ and $k_1 = -2$, so $-2p_1 + p_2 + p_3 = 0$, i.e. $p_3 = 2p_1 - p_2$.

S is linearly dependent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 1 + 2x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & -1 & 3 & 2 \\ 0 & 1 & -1 & 1 \end{array}\right] \xrightarrow{-R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & 2 & 1 \\ 0 & 1 & -1 & 1 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & -2 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 3$, i.e. $0 = 3$, which is impossible: the system is **inconsistent**. Hence u is not a linear combination of p_1, p_2, p_3.

$u \notin \operatorname{span}(S)$.

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side 0.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) = 0$.
a5	0.5	Conclusion **linearly dependent** with reason (zero row / free variable / nontrivial solution / det = 0). An explicit verified relation such as $p_3 = 2p_1 - p_2$ is a complete justification.
b1	0.5	System or augmented matrix with right-hand side $1 + 2x + x^2$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 3 \end{array}\right]$.
b3	0.5	States that the last row is impossible ($0 = 3$), hence $u \notin \operatorname{span}(S)$ (inconsistent: $0 = 3$). A correct argument via the dependence relation (e.g. that $\operatorname{span}(S) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathbf{u}$ is not a combination of two vectors) is also accepted.
Total	4	

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version E

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	The set $F(-\infty,\infty)$ of all real-valued functions defined on $(-\infty,\infty)$, with pointwise addition and scalar multiplication, is a vector space.	**TRUE**	Example 4.1.2 verifies all ten axioms for $F(-\infty,\infty)$. Source: Example 4.1.2 ($F(-\infty,\infty)$ is a vector space), line 10909
(b)	The set of all $n\times n$ matrices A with $\operatorname{tr}(A)=0$ is a subspace of M_{nn}.	**TRUE**	$\operatorname{tr}(O)=0$, $\operatorname{tr}(A+B)=\operatorname{tr}A+\operatorname{tr}B$, $\operatorname{tr}(kA)=k\operatorname{tr}A$ (Extra Problems 4.2.6). Source: Extra Problems 4.2.6 (trace zero), line 12157
(c)	If $\operatorname{span}(S)=\mathbb{R}^3$, then S must contain exactly three vectors.	**FALSE**	Extra Problems 4.3.2: the four vectors $(1,-1,0),(1,2,1),(1,0,1),(3,3,-1)$ span $\mathbb{R}^3$. Source: Extra Problems 4.3.2 (four vectors span $\mathbb{R}^3$), line 12992
(d)	If the Wronskian $W(x)$ of two functions satisfies $W(0)=0$, then the Wronskian test is inconclusive.	**FALSE**	Only $W(x)\equiv 0$ (zero for *every* x) makes the test inconclusive; e.g. $W(x)=x^2$ has $W(0)=0$ but $W(1)=1\neq 0$, so the functions are linearly independent (Remark, item 2). Source: Remark: Understanding the Wronskian Test, line 13547

Problem 2 – Multiple choice (1 mark each)

(a) Let $V=\mathbb{R}^3$ with the usual addition $(x_1,x_2,x_3)+(y_1,y_2,y_3)=(x_1+y_1,\ x_2+y_2,\ x_3+y_3)$ and the scalar multiplication $k(x_1,x_2,x_3)=(kx_1,\ x_2,\ x_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 9: $k(m\mathbf{u})=(km)\mathbf{u}$

B* Axiom 8: $(k+m)\mathbf{u}=k\mathbf{u}+m\mathbf{u}$

C Axiom 10: $1\mathbf{u}=\mathbf{u}$

D Axiom 2: $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$

Key: B (Axiom 8). Take $\mathbf{u}=(1,1,1)$, $k=m=1$: $(k+m)\mathbf{u}=2\mathbf{u}=(2,1,1)$ but $k\mathbf{u}+m\mathbf{u}=(1,1,1)+(1,1,1)=(2,2,2)$, so Axiom 8 fails. Axiom 9: $k(m\mathbf{u})=(kmu_1,u_2,u_3)=(km)\mathbf{u}$ holds; Axiom 10: $1\mathbf{u}=\mathbf{u}$ holds; Axiom 2 holds for the usual addition.

Source: Example 4.1.3 (NOT a vector space: $k(x_1,x_2)=(kx_1,0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1,x_2)=(kx_1,x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Example 4.1.4 extended to $\mathbb{R}^3$.

(b) Which of the following subsets W is a **subspace** of $V=F(-\infty,\infty)$ (with the usual operations)?

A $W=\{f : f(0)=1\}$ — **not a subspace.** The zero function has $f(0)=0\neq 1$, so $\mathbf{0}\notin W$.

B $W=\{f : f(x)\geq 0 \text{ for all } x\}$ — **not a subspace.** $f(x)=1$ is in W but $(-1)f=-1$ is not (condition (c) fails).

C* $W=C(-\infty,\infty)$, all continuous functions — **subspace.** Extra Problems 4.2.1: the zero function is continuous; sums and scalar multiples of continuous functions are continuous.

D $W=\{f : f(1)=2\}$ — **not a subspace.** The zero function has $f(1)=0\neq 2$, so $\mathbf{0}\notin W$.

Key: C. Source: Extra Problems 4.2.1 ($C(-\infty,\infty)$ is a subspace), line 11988; Example 4.1.2 ($F(-\infty,\infty)$ is a vector space), line 10909; Theorem 4.2.1 (The Subspace Test), line 11139. Adaptation: Extra Problems 4.2.1 as the subspace; non-subspaces built from the function-space examples.

(c) Let $\mathbf{v}_1=(1,1,0)$ and $\mathbf{v}_2=(0,1,1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\operatorname{span}\{\mathbf{v}_1,\mathbf{v}_2\}$?

A $(2, 3, 5)$

B* $(2, 5, 3)$

C $(2, 5, 4)$

D $(1, 1, 1)$

Key: B. Write $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ and match the candidate: two of the three components determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = 2$, $k_2 = 1$ would force the third component to be $1 \neq 5$. **B**: $k_1 = 2$, $k_2 = 3$ give third component $= 3$ ✓, so $(2, 5, 3) = 2\mathbf{v}_1 + 3\mathbf{v}_2$. C: $k_1 = 2$, $k_2 = 3$ would force the third component to be $3 \neq 4$. D: $k_1 = 1$, $k_2 = 0$ would force the third component to be $0 \neq 1$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = e^x$ and $f_2 = xe^x$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A* $W(x) = e^{2x}$; the set is linearly independent

B $W(x) = xe^{2x}$; the set is linearly independent

C $W(x) = xe^{2x}$; the set is linearly dependent because $W(0) = 0$

D $W(x) = e^{2x}$; the set is linearly dependent because xe^x is a multiple of e^x

Key: A. $(xe^x)' = e^x + xe^x$, so $W(x) = \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix} = e^{2x} + xe^{2x} - xe^{2x} = e^{2x} \neq 0$; linearly independent by Fact 4.4.2. xe^x is x times e^x, not a *scalar* multiple; xe^{2x} comes from forgetting the product rule.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, -1, 2), \qquad \mathbf{v}_2 = (1, 1, 0), \qquad \mathbf{v}_3 = (1, -3, 4),$$

and let $\mathbf{u} = (1, 2, 1)$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether $\mathbf{u}$ belongs to span(S), that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1, -1, 2) + k_2(1, 1, 0) + k_3(1, -3, 4) = (0, 0, 0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_2 + k_3 = 0 \\ -k_1 + k_2 - 3k_3 = 0 \\ 2k_1 + 4k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -1 & 1 & -3 & 0 \\ 2 & 0 & 4 & 0 \end{array}\right] \xrightarrow[-2R_1+R_3 \to R_3]{R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & -2 & 2 & 0 \end{array}\right] \xrightarrow{R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = 1$ and $k_1 = -2$, so $-2\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, i.e. $\mathbf{v}_3 = 2\mathbf{v}_1 - \mathbf{v}_2$.

S is linearly dependent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (1, 2, 1)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ -1 & 1 & -3 & 2 \\ 2 & 0 & 4 & 1 \end{array}\right] \xrightarrow[-2R_1+R_3\to R_3]{R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & -2 & 3 \\ 0 & -2 & 2 & -1 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & -2 & 3 \\ 0 & 0 & 0 & 2 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 2$, i.e. $0 = 2$, which is impossible: the system is **inconsistent**. Hence $\mathbf{u}$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

$\boxed{\mathbf{u} \notin \operatorname{span}(S).}$

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side $\mathbf{0}$.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) = 0$.
a5	0.5	Conclusion **linearly dependent** with reason (zero row / free variable / nontrivial solution / det = 0). An explicit verified relation such as $\mathbf{v}_3 = 2\mathbf{v}_1 - \mathbf{v}_2$ is a complete justification.
b1	0.5	System or augmented matrix with right-hand side $(1, 2, 1)$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 1 & 1 \\ 0 & 2 & -2 & 3 \\ 0 & 0 & 0 & 2 \end{array}\right]$.
b3	0.5	States that the last row is impossible ($0 = 2$), hence $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent: $0 = 2$). A correct argument via the dependence relation (e.g. that $\operatorname{span}(S) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathbf{u}$ is not a combination of two vectors) is also accepted.
Total	4	

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version F

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	To show that a set with two operations is a vector space, it is enough to verify Axiom 1 (closure under addition) and Axiom 6 (closure under scalar multiplication).	**FALSE**	Closure is only Step 3 of the procedure; Step 4 requires Axioms 2, 3, 4, 5, 7, 8, 9, 10 as well. Example 4.1.3 is closed under both operations yet fails Axiom 10. Source: Info box: To Show That a Set with Two Operations Is a Vector Space, line 10806; Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977
(b)	The first quadrant $W = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0\}$ is a subspace of $\mathbb{R}^2$.	**FALSE**	Not closed under scalar multiplication: $(-1)(5, 7) = (-5, -7) \notin W$ (Example 4.2.2). Source: Example 4.2.2 (First Quadrant Is NOT a Subspace), line 11200
(c)	$\text{span}\{1,\ x,\ x^2\} = P_2$.	**TRUE**	Remark 4.3.2(2): every $a + bx + cx^2$ is a linear combination of $1, x, x^2$. Source: Remark 4.3.2 (Important Spanning Sets), line 12664
(d)	If the Wronskian of $f_1, \ldots, f_n$ is not identically zero on $(-\infty, \infty)$, then the functions are linearly independent.	**TRUE**	Fact 4.4.2 (Wronskian test). Source: Fact 4.4.2 (Wronskian Test), line 13541

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1, x_2, x_3) + (y_1, y_2, y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1, x_2, x_3) = (kx_1,\ kx_2,\ 0)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 1: $\mathbf{u} + \mathbf{v}$ is in V (closure under addition)

B Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

C Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

D* Axiom 10: $1\mathbf{u} = \mathbf{u}$

Key: D (Axiom 10). Take $\mathbf{u} = (1, 2, 3)$: $1\mathbf{u} = (1, 2, 0) \neq \mathbf{u}$, so Axiom 10 fails. Axiom 1 holds (usual addition); Axiom 8: $(k + m)\mathbf{u} = ((k + m)u_1, (k + m)u_2, 0) = k\mathbf{u} + m\mathbf{u}$ holds; Axiom 9: $k(m\mathbf{u}) = (kmu_1, kmu_2, 0) = (km)\mathbf{u}$ holds.

Source: Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1, x_2) = (kx_1, x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Example 4.1.3 extended to $\mathbb{R}^3$.

(b) Which of the following subsets W is a **subspace** of $V = M_{22}$ (with the usual operations)?

A all matrices with $\det(A) = 0$ — **not a subspace.** Example 4.2.5: $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}$ have determinant 0 but $\det(A + B) = -7 \neq 0$.

B* all matrices with $\text{tr}(A) = 0$ — **subspace.** Extra Problems 4.2.6: $\text{tr}(O) = 0$, $\text{tr}(A+B) = \text{tr}\,A + \text{tr}\,B$, $\text{tr}(kA) = k\,\text{tr}\,A$.

C all matrices of the form $\begin{bmatrix} 1 & a \\ b & c \end{bmatrix}$ — **not a subspace.** $O \notin W$ because its $(1, 1)$ entry is $0 \neq 1$ (Extra Problems 4.2.3).

D all matrices whose four entries are all ≥ 0 — **not a subspace.** $I_2 \in W$ but $(-1)I_2 \notin W$ (condition (c) fails).

Key: B. Source: Extra Problems 4.2.6 (trace zero), line 12157; Example 4.2.5 (Singular Matrices Do NOT Form a Subspace), line 11317; Extra Problems 4.2.3 (matrices with $a_{11} = 1$), line 12077; Theorem 4.2.1 (The Subspace Test), line 11139. Adaptation: Notes' matrix examples; non-negative entries added as a non-subspace.

(c) Let $p_1 = 1 - x$ and $p_2 = 1 + x^2$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

A $1 + 2x - 3x^2$

B $3 + 2x + x^2$

C* $1 + 2x + 3x^2$

D $1 - 2x + 3x^2$

Key: C. Write $k_1p_1 + k_2p_2$ and match the candidate: two of the three coefficients determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = -2$, $k_2 = 3$ would force the x^2 coefficient to be $3 \neq -3$. B: $k_1 = -2$, $k_2 = 5$ would force the x^2 coefficient to be $5 \neq 1$. **C**: $k_1 = -2$, $k_2 = 3$ give x^2 coefficient $= 3$ ✓, so $1 + 2x + 3x^2 = -2p_1 + 3p_2$. D: $k_1 = 2$, $k_2 = -1$ would force the x^2 coefficient to be $-1 \neq 3$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = 1$ and $f_2 = x^2$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A $W(x) = 2x$; the set is linearly dependent because $W(0) = 0$

B $W(x) = -2x$; the set is linearly independent

C $W(x) = 2x$; the Wronskian test is inconclusive because $W(0) = 0$

D* $W(x) = 2x$; the set is linearly independent

Key: D. $W(x) = \begin{vmatrix} 1 & x^2 \\ 0 & 2x \end{vmatrix} = 2x$. $W(1) = 2 \neq 0$, so $W \not\equiv 0$ and the set is linearly independent (Fact 4.4.2). A zero at the single point $x = 0$ is irrelevant (Remark, item 2); $-2x$ is a sign error.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x^2, \qquad p_2 = x + x^2, \qquad p_3 = 1 - x + x^2,$$

and let $u = -1 + 4x + x^2$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether u belongs to $\text{span}(S)$, that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(1 + x^2) + k_2(x + x^2) + k_3(1 - x + x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_3 = 0 \\ k_2 - k_3 = 0 \\ k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right] \xrightarrow{-R_1+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = -1 + 4x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 1 & 1 & 1 & 1 \end{array}\right] \xrightarrow{-R_1+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 0 & 1 & 0 & 2 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

Back-substitution: $k_3 = -2 \Rightarrow k_3 = -2$; $k_2 - k_3 = 4 \Rightarrow k_2 = 2$; $k_1 + k_3 = -1 \Rightarrow k_1 = 1$. Hence $u = p_1 + 2p_2 - 2p_3$.

$u \in \text{span}(S)$: $u = p_1 + 2p_2 - 2p_3$.

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side 0.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) \neq 0$ computed correctly.
a5	0.5	Conclusion **linearly independent** with reason (three pivots / only the trivial solution / $\det \neq 0$).
b1	0.5	System or augmented matrix with right-hand side $-1 + 4x + x^2$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & -2 \end{array}\right]$.
b3	0.5	Correct coefficients and the combination written out: $u = p_1 + 2p_2 - 2p_3$. Cramer's rule / inverse matrix accepted.
Total	4	

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version G

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	In a vector space V, $(-1)\mathbf{u} = -\mathbf{u}$ for every vector $\mathbf{u}$ in V.	**TRUE**	This is Theorem 4.1.1(c). Source: Theorem 4.1.1, line 10790
(b)	The set $W = \{a + bx + cx^2 : a, b, c \text{ are integers}\}$ is a subspace of P_2.	**FALSE**	Not closed under scalar multiplication: $\frac{1}{2}(1 + x + x^2) \notin W$ (Extra Problems 4.2.4). Source: Extra Problems 4.2.4 (integer coefficients), line 12098
(c)	The zero vector belongs to span(S) for every nonempty set S of vectors in V.	**TRUE**	$\mathbf{0} = 0\mathbf{w}_1 + \cdots + 0\mathbf{w}_r$; also span$(S)$ is a subspace (Theorem 4.3.1). Source: Definition 4.3.1 (Span), line 12614; Theorem 4.3.1 (Span Is a Subspace), line 12627
(d)	If the Wronskian of $f_1, \ldots, f_n$ is identically zero on $(-\infty, \infty)$, then the functions are linearly dependent.	**FALSE**	When $W(x) \equiv 0$ the Wronskian test is inconclusive (Remark after Fact 4.4.2, item 5). Source: Remark: Understanding the Wronskian Test, line 13547; Fact 4.4.2 (Wronskian Test), line 13541

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (kx_1,\ k^2x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 10: $1\mathbf{u} = \mathbf{u}$

B Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

C* Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

D Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

Key: C (Axiom 8). Take $\mathbf{u} = (1, 1)$, $k = m = 1$: $(k + m)\mathbf{u} = 2(1, 1) = (2, 4)$ but $k\mathbf{u} + m\mathbf{u} = (1, 1) + (1, 1) = (2, 2)$, so Axiom 8 fails. Axiom 10: $1\mathbf{u} = (u_1, u_2)$ holds; Axiom 9: $k(m\mathbf{u}) = (kmu_1, k^2m^2u_2) = (km)\mathbf{u}$ holds; Axiom 7: $k(\mathbf{u} + \mathbf{v}) = (k(u_1 + v_1), k^2(u_2 + v_2)) = k\mathbf{u} + k\mathbf{v}$ holds.

Source: Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1, x_2) = (kx_1, x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: New scalar multiplication in the style of Examples 4.1.3-4.1.4; only Axiom 8 fails.

(b) Which of the following subsets of $V = \mathbb{R}^2$ (with the usual operations) is **not** a subspace of V?

A $W = \{(x, y) : x = 0\}$ (the y-axis) — **subspace.** Example 4.2.9: a line through the origin; $(0, y_1) + (0, y_2) = (0, y_1 + y_2)$ and $k(0, y) = (0, ky)$.

B $W = \{(x, y) : 2x - 5y = 0\}$ — **subspace.** Solution space of a homogeneous equation (Example 4.2.7).

C $W = \{(0, 0)\}$ — **subspace.** The zero subspace (Example 4.2.1).

D* $W = \{(x, y) : xy \geq 0\}$ — **not a subspace.** $(1, 0) \in W$ and $(0, -1) \in W$, but their sum $(1, -1)$ has $xy = -1 < 0$ (condition (b) fails).

Key: D. Source: Example 4.2.9 (Union of Subspaces Is NOT a Subspace), line 11687; Example 4.2.7 (Solution Spaces of Homogeneous Systems), line 11477; Example 4.2.1 (The Zero Subspace), line 11169; Remark 4.2.1 (Showing a Set Is NOT a Subspace), line 11154. Adaptation: The set $xy \geq 0$ (closed under scalars but not sums) tests condition (b) as in Remark 4.2.1.

(c) Let $\mathbf{v}_1 = (1, 0, 0)$ and $\mathbf{v}_2 = (0, 1, 2)$ in $\mathbb{R}^3$. Which of the following vectors belongs to span$\{\mathbf{v}_1, \mathbf{v}_2\}$?

A $(4, 2, 1)$

B $(1, 2, 3)$

C $(0, 1, 3)$

D* $(4, 1, 2)$

Key: D. Write $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ and match the candidate: two of the three components determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = 4$, $k_2 = 2$ would force the third component to be $4 \neq 1$. B: $k_1 = 1$, $k_2 = 2$ would force the third component to be $4 \neq 3$. C: $k_1 = 0$, $k_2 = 1$ would force the third component to be $2 \neq 3$. **D**: $k_1 = 4$, $k_2 = 1$ give third component $= 2$ ✓, so $(4, 1, 2) = 4\mathbf{v}_1 + \mathbf{v}_2$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = e^{2x}$ and $f_2 = e^{3x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A $W(x) = 5e^{5x}$; the set is linearly independent

B* $W(x) = e^{5x}$; the set is linearly independent

C $W(x) = e^{5x}$; the set is linearly dependent because $e^{3x} = e^x \cdot e^{2x}$

D $W(x) = e^{6x}$; the set is linearly independent

Key: B. $W(x) = \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix} = 3e^{5x} - 2e^{5x} = e^{5x} \neq 0$; linearly independent (Fact 4.4.2). $5e^{5x}$ adds instead of subtracting; e^{3x} is not a scalar multiple of e^{2x}; e^{6x} multiplies the functions instead of computing the determinant.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 2, 1), \qquad \mathbf{v}_2 = (2, 1, -1), \qquad \mathbf{v}_3 = (1, 1, 2),$$

and let $\mathbf{u} = (4, 3, -3)$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1, 2, 1) + k_2(2, 1, -1) + k_3(1, 1, 2) = (0, 0, 0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + 2k_2 + k_3 = 0 \\ 2k_1 + k_2 + k_3 = 0 \\ k_1 - k_2 + 2k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & -1 & 2 & 0 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & -3 & 1 & 0 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (4, 3, -3)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & -1 & 2 & -3 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -1 & -5 \\ 0 & -3 & 1 & -7 \end{array}\right] \xrightarrow{-R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -1 & -5 \\ 0 & 0 & 2 & -2 \end{array}\right]$$

Back-substitution: $2k_3 = -2 \Rightarrow k_3 = -1$; $-3k_2 - k_3 = -5 \Rightarrow k_2 = 2$; $k_1 + 2k_2 + k_3 = 4 \Rightarrow k_1 = 1$. Hence $\mathbf{u} = \mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3$.

$\mathbf{u} \in \text{span}(S)$: $\mathbf{u} = \mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3$.

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side $\mathbf{0}$.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) \neq 0$ computed correctly.
a5	0.5	Conclusion **linearly independent** with reason (three pivots / only the trivial solution / det $\neq 0$).
b1	0.5	System or augmented matrix with right-hand side $(4, 3, -3)$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 2 & 1 & 4 \\ 0 & -3 & -1 & -5 \\ 0 & 0 & 2 & -2 \end{array}\right]$.
b3	0.5	Correct coefficients and the combination written out: $\mathbf{u} = \mathbf{v}_1 + 2\mathbf{v}_2 - \mathbf{v}_3$. Cramer's rule / inverse matrix accepted.
Total	4	

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version H

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	If $\mathbf{u}$ is a vector in a vector space and k is a scalar with $k\mathbf{u} = \mathbf{0}$, then necessarily $k = 0$.	**FALSE**	Theorem 4.1.1(d) says $k = 0$ *or* $\mathbf{u} = \mathbf{0}$. Counterexample: $k = 5$ and $\mathbf{u} = \mathbf{0}$ give $5\mathbf{0} = \mathbf{0}$ with $k \neq 0$ (Theorem 4.1.1(b)). Source: Theorem 4.1.1, line 10790
(b)	The set $W = \{a + bx + cx^2 : a + b + c = 0\}$ is a subspace of P_2.	**TRUE**	The zero polynomial has $0+0+0 = 0$; the condition is preserved under sums and scalar multiples (Extra Problems 4.2.2). Source: Extra Problems 4.2.2 ($a + b + c = 0$ in P_2), line 12017
(c)	The set $S = \{(1,-1,0),\ (1,2,1),\ (1,0,1),\ (3,3,-1)\}$ cannot span $\mathbb{R}^3$ because it contains more than three vectors.	**FALSE**	Extra Problems 4.3.2 row reduces to three pivots: the system is consistent for every (a,b,c), so $\text{span}(S) = \mathbb{R}^3$. Source: Extra Problems 4.3.2 (four vectors span $\mathbb{R}^3$), line 12992
(d)	Every set of five matrices in M_{22} is linearly dependent.	**TRUE**	Fact 4.4.1(3) with $r = 5 > 2 \cdot 2 = 4$. Source: Fact 4.4.1 (Dimension Argument for Linear Dependence), line 13452

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1,x_2,x_3) + (y_1,y_2,y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1,x_2,x_3) = (0,\ kx_2,\ kx_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A* Axiom 10: $1\mathbf{u} = \mathbf{u}$

B Axiom 8: $(k+m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

C Axiom 4: there is a zero vector $\mathbf{0}$ with $\mathbf{0} + \mathbf{u} = \mathbf{u}$

D Axiom 1: $\mathbf{u} + \mathbf{v}$ is in V (closure under addition)

Key: A (Axiom 10). Take $\mathbf{u} = (5,1,2)$: $1\mathbf{u} = (0,1,2) \neq \mathbf{u}$, so Axiom 10 fails. Axiom 8: $(k+m)\mathbf{u} = (0,(k+m)u_2,(k+m)u_3) = k\mathbf{u} + m\mathbf{u}$ holds; Axioms 4 and 1 hold for the usual addition.

Source: Example 4.1.3 (NOT a vector space: $k(x_1,x_2) = (kx_1,0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1,x_2) = (kx_1,x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Mirror image of Example 4.1.3 in $\mathbb{R}^3$.

(b) Which of the following subsets of $V = F(-\infty,\infty)$ (with the usual operations) is **not** a subspace of V?

A* $W = \{f : f(0) = 1\}$ — **not a subspace.** The zero function has $f(0) = 0 \neq 1$, so $\mathbf{0} \notin W$.

B $W = \{f : f(0) = 0\}$ — **subspace.** $\mathbf{0}(0) = 0$; $(f+g)(0) = 0+0 = 0$; $(kf)(0) = k \cdot 0 = 0$.

C $W = C^1(-\infty,\infty)$, all continuously differentiable functions — **subspace.** Extra Problems 4.2.5.

D $W = P_2$, all polynomials of degree at most 2 — **subspace.** P_2 is a vector space under the same operations (Definition of P_n, hierarchy of function spaces), hence a subspace of $F(-\infty,\infty)$.

Key: A. Source: Extra Problems 4.2.5 ($C^1(-\infty,\infty)$), line 12128; Definition box: The Vector Space P_n / Hierarchy of Function Spaces, line 11369; Theorem 4.2.1 (The Subspace Test), line 11139. Adaptation: Function-space hierarchy from the notes; $f(0) = 1$ added as the non-subspace.

(c) Let $p_1 = x + x^2$ and $p_2 = 1$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

A* $5 + 2x + 2x^2$

B $5 + 2x + 5x^2$

C $2 + 5x + 2x^2$

D $5 + 2x - 2x^2$

Key: A. Write $k_1p_1 + k_2p_2$ and match the candidate: two of the three coefficients determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). **A**: $k_1 = 2$, $k_2 = 5$ give x^2 coefficient $= 2$ ✓, so $5 + 2x + 2x^2 = 2p_1 + 5p_2$. B: $k_1 = 2$, $k_2 = 5$ would force the x^2 coefficient to be $2 \neq 5$. C: $k_1 = 5$, $k_2 = 2$ would force the x^2 coefficient to be $5 \neq 2$. D: $k_1 = 2$, $k_2 = 5$ would force the x^2 coefficient to be $2 \neq -2$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = x$ and $f_2 = 1 + x$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A $W(x) = 1$; the set is linearly independent

B $W(x) = -1$; the set is linearly dependent because $W(x)$ is negative

C* $W(x) = -1$; the set is linearly independent

D $W(x) = 0$ for all x; the Wronskian test is inconclusive

Key: C. $W(x) = \begin{vmatrix} x & 1+x \\ 1 & 1 \end{vmatrix} = x - (1 + x) = -1 \neq 0$ for every x, so the set is linearly independent (Fact 4.4.2). The sign of W does not matter; $W = 1$ is a sign error; W is not the zero function.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 2 + x - x^2, \qquad p_2 = 1 + x + x^2, \qquad p_3 = 4 + 3x + x^2,$$

and let $u = 1 + 2x + x^2$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(2 + x - x^2) + k_2(1 + x + x^2) + k_3(4 + 3x + x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} 2k_1 + k_2 + 4k_3 = 0 \\ k_1 + k_2 + 3k_3 = 0 \\ -k_1 + k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 2 & 1 & 4 & 0 \\ 1 & 1 & 3 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & 1 & 4 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right] \xrightarrow[R_1+R_3 \to R_3]{-2R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 2 & 4 & 0 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = -2$ and $k_1 = -1$, so $-p_1 - 2p_2 + p_3 = 0$, i.e. $p_3 = p_1 + 2p_2$.

S is linearly dependent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 1 + 2x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 2 & 1 & 4 & 1 \\ 1 & 1 & 3 & 2 \\ -1 & 1 & 1 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 2 & 1 & 4 & 1 \\ -1 & 1 & 1 & 1 \end{array}\right] \xrightarrow[R_1+R_3 \to R_3]{-2R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & -1 & -2 & -3 \\ 0 & 2 & 4 & 3 \end{array}\right]$$

$$\xrightarrow{2R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & -3 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = -3$, i.e. $0 = -3$, which is impossible: the system is **inconsistent**. Hence u is not a linear combination of p_1, p_2, p_3.

$u \notin \operatorname{span}(S)$.

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side 0.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 3 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) = 0$.
a5	0.5	Conclusion **linearly dependent** with reason (zero row / free variable / nontrivial solution / det = 0). An explicit verified relation such as $p_3 = p_1 + 2p_2$ is a complete justification.
b1	0.5	System or augmented matrix with right-hand side $1 + 2x + x^2$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 1 & 3 & 2 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & -3 \end{array}\right]$.
b3	0.5	States that the last row is impossible ($0 = -3$), hence $u \notin \operatorname{span}(S)$ (inconsistent: $0 = -3$). A correct argument via the dependence relation (e.g. that $\operatorname{span}(S) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathbf{u}$ is not a combination of two vectors) is also accepted.
Total	4	

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version I

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	In a vector space V, the equation $k\mathbf{0} = \mathbf{0}$ holds only when the scalar k equals 0.	**FALSE**	Theorem 4.1.1(b): $k\mathbf{0} = \mathbf{0}$ for *every* scalar k. Source: Theorem 4.1.1, line 10790
(b)	The set $C(-\infty, \infty)$ of all continuous functions on $(-\infty, \infty)$ is a subspace of $F(-\infty, \infty)$.	**TRUE**	The zero function is continuous, and sums and scalar multiples of continuous functions are continuous (Extra Problems 4.2.1). Source: Extra Problems 4.2.1 ($C(-\infty, \infty)$ is a subspace), line 11988
(c)	If $\mathbf{u} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$, then $\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ for some scalars k_1, k_2.	**TRUE**	Definition 4.3.1: the span is the set of all linear combinations. Source: Definition 4.3.1 (Span), line 12614; Definition 4.2.4 (Linear Combination), line 11760
(d)	Every set of two nonzero vectors in $\mathbb{R}^3$ is linearly independent.	**FALSE**	Theorem 4.4.1(c): $(1,0,0)$ and $(2,0,0)$ are nonzero but one is a scalar multiple of the other, so the set is linearly dependent. Source: Theorem 4.4.1, line 13253

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (x_1,\ kx_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

B Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

C* Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

D Axiom 10: $1\mathbf{u} = \mathbf{u}$

Key: C (Axiom 8). Take $\mathbf{u} = (5, 2)$, $k = 2$, $m = 3$: $(k + m)\mathbf{u} = 5(5, 2) = (5, 10)$ but $k\mathbf{u} + m\mathbf{u} = (5, 4) + (5, 6) = (10, 10)$, so Axiom 8 fails. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = (u_1 + v_1, k(u_2 + v_2)) = k\mathbf{u} + k\mathbf{v}$ holds; Axiom 9: $k(m\mathbf{u}) = (u_1, kmu_2) = (km)\mathbf{u}$ holds; Axiom 10: $1\mathbf{u} = \mathbf{u}$ holds.

Source: Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1, x_2) = (kx_1, x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Mirror image of Example 4.1.4.

(b) Which of the following subsets W is a **subspace** of $V = \mathbb{R}^3$ (with the usual operations)?

A $W = \{(x, y, z) : x \geq 0\}$ — **not a subspace.** $(1, 0, 0) \in W$ but $(-1)(1, 0, 0) = (-1, 0, 0) \notin W$ (condition (c) fails).

B $W = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$ — **not a subspace.** $(1, 0, 0) \in W$ but $2(1, 0, 0) = (2, 0, 0) \notin W$ since $4 > 1$ (condition (c) fails).

C $W = \{(x, y, z) : xyz = 0\}$ — **not a subspace.** $(1, 1, 0) \in W$ and $(0, 0, 1) \in W$, but $(1, 1, 0) + (0, 0, 1) = (1, 1, 1)$ has $xyz = 1 \neq 0$ (condition (b) fails).

D* $W = \{(x, y, z) : z = x - 2y\}$ — **subspace.** Solution space of the homogeneous equation $x - 2y - z = 0$ (Example 4.2.7).

Key: D. Source: Example 4.2.7 (Solution Spaces of Homogeneous Systems), line 11477; Remark 4.2.1 (Showing a Set Is NOT a Subspace), line 11154; Example 4.2.2 (First Quadrant Is NOT a Subspace), line 11200. Adaptation: Homogeneous plane as the subspace; three non-subspaces built with Remark 4.2.1 counterexamples.

(c) Let $\mathbf{v}_1 = (1, 0, 1)$ and $\mathbf{v}_2 = (0, 1, 1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

A $(3, -1, 4)$

B $(3, 1, 2)$

C* $(3, -1, 2)$

D $(1, 1, 1)$

Key: C. Write $k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ and match the candidate: two of the three components determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = 3, k_2 = -1$ would force the third component to be $2 \neq 4$. B: $k_1 = 3, k_2 = 1$ would force the third component to be $4 \neq 2$. **C**: $k_1 = 3, k_2 = -1$ give third component $= 2$ ✓, so $(3, -1, 2) = 3\mathbf{v}_1 - \mathbf{v}_2$. D: $k_1 = 1, k_2 = 1$ would force the third component to be $2 \neq 1$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = x^2$ and $f_2 = x^3$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A $W(x) = x^4$; the set is linearly dependent because $W(0) = 0$

B* $W(x) = x^4$; the set is linearly independent

C $W(x) = 5x^4$; the set is linearly independent

D $W(x) = 0$ for all x; the Wronskian test is inconclusive

Key: B. $W(x) = \begin{vmatrix} x^2 & x^3 \\ 2x & 3x^2 \end{vmatrix} = 3x^4 - 2x^4 = x^4$. $W(1) = 1 \neq 0$, so $W \not\equiv 0$ and the set is linearly independent (Fact 4.4.2). $W(0) = 0$ at one point does not give dependence (Remark, item 2); $5x^4$ adds instead of subtracting.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, -2, 1), \qquad \mathbf{v}_2 = (3, 1, 2), \qquad \mathbf{v}_3 = (5, -3, 4),$$

and let $\mathbf{u} = (2, 1, 3)$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether $\mathbf{u}$ belongs to span(S), that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{0}$, i.e. $k_1(1, -2, 1) + k_2(3, 1, 2) + k_3(5, -3, 4) = (0, 0, 0)$. Comparing components gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + 3k_2 + 5k_3 = 0 \\ -2k_1 + k_2 - 3k_3 = 0 \\ k_1 + 2k_2 + 4k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ -2 & 1 & -3 & 0 \\ 1 & 2 & 4 & 0 \end{array}\right] \xrightarrow[-R_1+R_3 \to R_3]{2R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ 0 & 7 & 7 & 0 \\ 0 & -1 & -1 & 0 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 7 & 7 & 0 \end{array}\right]$$

$$\xrightarrow{7R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The coefficient matrix has only two pivots: the third row is a zero row, so k_3 is a free variable and the homogeneous system has **nontrivial solutions**. For instance $k_3 = 1$ gives $k_2 = -1$ and $k_1 = -2$, so $-2\mathbf{v}_1 - \mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$, i.e. $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$.

S is linearly dependent.

(b) Set $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$ with $\mathbf{u} = (2, 1, 3)$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ -2 & 1 & -3 & 1 \\ 1 & 2 & 4 & 3 \end{array}\right] \xrightarrow[-R_1+R_3\to R_3]{2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & 7 & 7 & 5 \\ 0 & -1 & -1 & 1 \end{array}\right] \xrightarrow{R_2\leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & -1 & -1 & 1 \\ 0 & 7 & 7 & 5 \end{array}\right]$$

$$\xrightarrow{7R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 3 & 5 & 2 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 12 \end{array}\right]$$

The last row reads $0k_1 + 0k_2 + 0k_3 = 12$, i.e. $0 = 12$, which is impossible: the system is **inconsistent**. Hence $\mathbf{u}$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$.

$\mathbf{u} \notin \operatorname{span}(S)$.

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side $\mathbf{0}$.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 3 & 5 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) = 0$.
a5	0.5	Conclusion **linearly dependent** with reason (zero row / free variable / nontrivial solution / det = 0). An explicit verified relation such as $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$ is a complete justification.
b1	0.5	System or augmented matrix with right-hand side $(2, 1, 3)$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 3 & 5 & 2 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & 0 & 12 \end{array}\right]$.
b3	0.5	States that the last row is impossible ($0 = 12$), hence $\mathbf{u} \notin \operatorname{span}(S)$ (inconsistent: $0 = 12$). A correct argument via the dependence relation (e.g. that $\operatorname{span}(S) = \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ and $\mathbf{u}$ is not a combination of two vectors) is also accepted.
Total	4	

Source: Example 4.4.1 (Linear Dependence in $\mathbb{R}^3$), line 13280; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.

Version J

Problem 1 – True/False (0.5 mark each)

	Statement	Key	Reason / source
(a)	In a real vector space, the scalars may be any real numbers, including irrational numbers such as $\sqrt{2}$.	**TRUE**	Scalars are real numbers (Definition 4.1.1); the notes use $k = \sqrt{2}$ in Extra Problems 4.2.9(c). Source: Definition 4.1.1 (Vector Space Axioms), line 10737; Extra Problems 4.2.9 (subspaces of P_3), line 12451
(b)	If A is an $m \times n$ matrix and $\mathbf{b} \neq \mathbf{0}$, then the solution set of $A\mathbf{x} = \mathbf{b}$ is a subspace of $\mathbb{R}^n$.	**FALSE**	$A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$, so the zero vector is not a solution; condition (a) fails (Example 4.2.8). Source: Example 4.2.8 (Solution Set of a Non-Homogeneous System), line 11525
(c)	The polynomial x^2 belongs to span$\{1 + x^2,\ x\}$.	**FALSE**	$k_1(1 + x^2) + k_2x = k_1 + k_2x + k_1x^2$; matching x^2 requires $k_1 = 0$ (constant term) and $k_1 = 1$ (x^2 term), impossible. Source: Definition 4.3.1 (Span), line 12614; Example 4.2.12 (Linear Combination of Polynomials), line 11899
(d)	A set consisting of a single nonzero vector is linearly independent.	**TRUE**	Theorem 4.4.1(b). Source: Theorem 4.4.1, line 13253

Problem 2 – Multiple choice (1 mark each)

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1, x_2, x_3) + (y_1, y_2, y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1, x_2, x_3) = (x_1,\ x_2,\ kx_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

A Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

B* Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

C Axiom 10: $1\mathbf{u} = \mathbf{u}$

D Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

Key: B (Axiom 8). Take $\mathbf{u} = (1, 1, 1)$, $k = m = 1$: $(k + m)\mathbf{u} = 2\mathbf{u} = (1, 1, 2)$ but $k\mathbf{u} + m\mathbf{u} = (1, 1, 1) + (1, 1, 1) = (2, 2, 2)$, so Axiom 8 fails. Axiom 6 holds since $(u_1, u_2, ku_3) \in \mathbb{R}^3$; Axiom 10: $1\mathbf{u} = \mathbf{u}$ holds; Axiom 9: $k(m\mathbf{u}) = (u_1, u_2, kmu_3) = (km)\mathbf{u}$ holds.

Source: Example 4.1.3 (NOT a vector space: $k(x_1, x_2) = (kx_1, 0)$), line 10977; Example 4.1.4 (NOT a vector space: $k(x_1, x_2) = (kx_1, x_2)$), line 11026; Definition 4.1.1 (Vector Space Axioms), line 10737. Adaptation: Example 4.1.4 extended to $\mathbb{R}^3$ (modified third component).

(b) Which of the following subsets of $V = P_2$ (with the usual operations) is **not** a subspace of V?

A $W = \{p \in P_2 : p(0) = 0\}$ — **subspace.** $p(0) = a$, so $W = \{bx + cx^2\}$: the zero polynomial is in W and sums/scalar multiples keep $a = 0$ (compare Extra Problems 4.2.9(a)).

B* $W = \{a + bx + cx^2 : a + b + c = 2\}$ — **not a subspace.** The zero polynomial has coefficient sum $0 \neq 2$, so $\mathbf{0} \notin W$.

C $W = \{a + bx + cx^2 : a = 2c\}$ — **subspace.** $0 = 2 \cdot 0$; if $a_1 = 2c_1$ and $a_2 = 2c_2$ then $a_1 + a_2 = 2(c_1 + c_2)$ and $ka_1 = 2(kc_1)$.

D $W = \{p \in P_2 : p(1) = 0\}$ — **subspace.** $p(1) = a + b + c$, so this is the subspace of Extra Problems 4.2.2.

Key: B. Source: Extra Problems 4.2.2 ($a + b + c = 0$ in P_2), line 12017; Extra Problems 4.2.9 (subspaces of P_3), line 12451; Example 4.2.6 (Degree Exactly 2 Is NOT a Subspace), line 11424; Theorem 4.2.1 (The Subspace Test), line 11139. Adaptation: Homogeneous coefficient conditions (subspaces) versus a non-homogeneous one.

(c) Let $p_1 = 1 + 2x$ and $p_2 = x^2$ in P_2. Which of the following polynomials belongs to span$\{p_1, p_2\}$?

A $2 + 2x - x^2$

B* $2 + 4x - x^2$

C $4 + 2x - x^2$

D $1 + 4x - x^2$

Key: B. Write $k_1p_1 + k_2p_2$ and match the candidate: two of the three coefficients determine k_1, k_2 uniquely, and the remaining one must agree (Examples 4.2.10–4.2.11 method). A: $k_1 = 2$, $k_2 = -1$ would force the x coefficient to be $4 \neq 2$. **B**: $k_1 = 2$, $k_2 = -1$ give x coefficient $= 4$ ✓, so $2 + 4x - x^2 = 2p_1 - p_2$. C: $k_1 = 4$, $k_2 = -1$ would force the x coefficient to be $8 \neq 2$. D: $k_1 = 1$, $k_2 = -1$ would force the x coefficient to be $2 \neq 4$.

Source: Definition 4.3.1 (Span), line 12614; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Example 4.2.12 (Linear Combination of Polynomials), line 11899. Adaptation: new generators and candidates; same method as the notes.

(d) Let $f_1 = e^{-x}$ and $f_2 = e^{x}$ be functions in $C^{\infty}(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

A* $W(x) = 2$; the set is linearly independent

B $W(x) = 0$ for all x; the Wronskian test is inconclusive

C $W(x) = -2$; the set is linearly independent

D $W(x) = 2$; the set is linearly dependent because $W(x)$ is constant

Key: A. $W(x) = \begin{vmatrix} e^{-x} & e^{x} \\ -e^{-x} & e^{x} \end{vmatrix} = e^{-x}e^{x} - e^{x}(-e^{-x}) = 1 + 1 = 2 \neq 0$, so the set is linearly independent (Fact 4.4.2). $W = 0$ comes from the sign error $1 - 1$; -2 is a sign error; a constant nonzero Wronskian is exactly what Fact 4.4.2 needs.

Source: Definition 4.4.2 (Wronskian), line 13526; Fact 4.4.2 (Wronskian Test), line 13541; Remark: Understanding the Wronskian Test, line 13547; Example 4.4.5 (Wronskian with Exponentials), line 13617; Example 4.4.7 (Wronskian with Scalar Multiples), line 13707. Adaptation: new pair of functions (2×2 Wronskian); same test and interpretation as the notes.

Problem 3 – Written (4 marks)

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + 2x, \qquad p_2 = x - x^2, \qquad p_3 = 1 + x^2,$$

and let $u = 1 - x + x^2$.

(a) [2.5] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why.

Model solution.

(a) Set $k_1p_1 + k_2p_2 + k_3p_3 = 0$, i.e. $k_1(1 + 2x) + k_2(x - x^2) + k_3(1 + x^2) = 0 + 0x + 0x^2$. Comparing coefficients of 1, x and x^2 gives the homogeneous system and its augmented matrix:

$$\begin{cases} k_1 + k_3 = 0 \\ 2k_1 + k_2 = 0 \\ -k_2 + k_3 = 0 \end{cases} \qquad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right] \xrightarrow{-2R_1+R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 \end{array}\right] \xrightarrow{R_2+R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right]$$

There are three pivots and no free variables, so the homogeneous system has **only the trivial solution** $k_1 = k_2 = k_3 = 0$.

S is linearly independent.

(b) Set $k_1p_1 + k_2p_2 + k_3p_3 = u$ with $u = 1 - x + x^2$. The coefficient matrix is the same, so the same row operations apply to the new right-hand side:

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & 1 & 0 & -1 \\ 0 & -1 & 1 & 1 \end{array}\right] \xrightarrow{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -3 \\ 0 & -1 & 1 & 1 \end{array}\right] \xrightarrow{R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & -1 & -2 \end{array}\right]$$

Back-substitution: $-k_3 = -2 \Rightarrow k_3 = 2$; $k_2 - 2k_3 = -3 \Rightarrow k_2 = 1$; $k_1 + k_3 = 1 \Rightarrow k_1 = -1$. Hence $u = -p_1 + p_2 + 2p_3$.

$u \in \text{span}(S)$**:** $u = -p_1 + p_2 + 2p_3$**.**

Version-specific marking guide.

Check	Marks	What to look for in this version
a1	0.5	Vector equation / coefficient matching set up with unknowns k_1, k_2, k_3 and right-hand side 0.
a2	0.5	The system shown in the model solution (or its augmented matrix). Row/column transposition accepted.
a3	0.5	Zeros below the first pivot with correct operations (any valid pivot choice).
a4	0.5	A correct row echelon form, e.g. $\left[\begin{array}{ccc\|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right]$ (any row-equivalent REF/RREF accepted); or $\det(A) \neq 0$ computed correctly.
a5	0.5	Conclusion **linearly independent** with reason (three pivots / only the trivial solution / $\det \neq 0$).
b1	0.5	System or augmented matrix with right-hand side $1 - x + x^2$ (re-using the matrix of (a) is fine).
b2	0.5	Correct REF of $[A \mid \mathbf{u}]$, e.g. $\left[\begin{array}{ccc\|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & -1 & -2 \end{array}\right]$.
b3	0.5	Correct coefficients and the combination written out: $u = -p_1 + p_2 + 2p_3$. Cramer's rule / inverse matrix accepted.
Total	4	

Source: Example 4.4.2 (Linear Independence in P_3), line 13357; Example 4.2.10 (Vector IS a Linear Combination), line 11774; Example 4.2.11 (Vector Is NOT a Linear Combination), line 11843; Definition 4.4.1 (Linear Independence and Dependence), line 13236. Adaptation: new numbers; the structure (set up, system, reduce, interpret) is exactly the taught method.